Original Article

Trends in breast cancer incidence rates by race/ethnicity: Patterns by stage, socioeconomic position, and geography in the United States, 1999-2017

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BACKGROUND: The incidence rate of breast cancer has been increasing over time across race/ethnicity in the United States. It is unclear whether these trends differ among stage, poverty, and geography subgroups. **METHODS:** Using data from the North American Association of Central Cancer Registries, this study estimated trends in age-adjusted breast cancer incidence rates among women aged 50 to 84 years from 1999 to 2017 by race/ethnicity (non-Hispanic Black, non-Hispanic White, and Hispanic) and across subgroups (stage, county-level poverty, county urban/rural status, and geographic region [West, Midwest, South, and Northeast]). **RESULTS:** From 2004 to 2017, breast cancer incidence rates increased across race/ethnicity and subgroups, with the greatest average annual percent increases observed for non-Hispanic Black women, overall (0.9%) and those living in lower poverty areas (0.8%), rural areas (1.2%), and all regions except the West (0.8%-1.0%). Stronger increases among non-Hispanic Black women were observed for local-stage disease and for some subgroups of distant-stage disease. Non-Hispanic Black women had the smallest decrease in regional-stage disease across most subgroups. Similarly, Hispanic women had the strongest increases in some subgroups, including areas with higher poverty (0.6%-1.2%) and in the West (0.8%), for local- and distant-stage disease. **CONCLUSIONS:** These trends highlight concerns for an increasing burden of breast cancer among subpopulations, with some already experiencing disparate breast cancer mortality rates, and they highlight the need for targeted breast cancer prevention and efforts to reduce mortality disparities in areas with increasing incidence. ***Cancer* 2022;128:1015-1023**. *© 2021 American Cancer Society*.

KEYWORDS: breast, cancer, ethnicity, geography, race, socioeconomic, trends.

INTRODUCTION

The incidence rate of breast cancer has been changing over time across racial/ethnic groups in the United States. Non-Hispanic Black women have historically had lower rates of breast cancer than non-Hispanic White women, but recent studies suggest trends toward convergence of the breast cancer incidence rates between Black and White women, especially for women aged 50 years or older.[1-3] According to US data from the North American Association of Central Cancer Registries (NAACCR), the age-adjusted incidence rate of breast cancer among women age 50 years or older from 2013 to 2017 was 338.9 per 100,000 for non-Hispanic Black women, 353.4 per 100,000 for non-Hispanic White women, and 253.5 per 100,000 for Hispanic women.[4] According to data from the Surveillance, Epidemiology, and End Results (SEER) program, the incidence rate of breast cancer among women aged 50 years or older increased by 0.9% per year among non-Hispanic Black women, by 0.3% per year among non-Hispanic White women, and by 0.5% per year among Hispanic women from 2008 to 2017.[5]

Despite having a lower breast cancer incidence rate, non-Hispanic Black women overall have a 40% higher breast cancer mortality rate than non-Hispanic White women.[5] This racial disparity in breast cancer mortality has persisted for 30 years even with advances in breast cancer therapeutics, and it has been stable since 2011 because of the slowed decline in mortality among non-Hispanic White women.[6] Similarly to Black women, Hispanic women have a lower incidence of breast cancer and lower survival from breast cancer in comparison with non-Hispanic White women.[6]

Increasing breast cancer rates are concerning for existing disparities in breast cancer mortality. There may be additional variation in breast cancer incidence trends by race/ethnicity concealed in important subgroups where additional disparities have been observed, such as socioeconomic position and geography subgroups. The breast cancer survival disparity between Black and White women persists after adjustments for socioeconomic position using measures such as

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individual- and area-level income and education.[7,8] Lower survival has also been observed among women living in areas with greater poverty within racial/ethnic groups.[9,10] There is variation in breast cancer mortality rates and in the Black-White breast cancer mortality disparity by geography within the United States.[5,6] Furthermore, NAACCR covers almost 99% of the US population,[11] whereas SEER covers up to 48% of the US population.[12] Thus, examining trends in breast cancer incidence by race/ethnicity across additional groups in NAACCR may reveal additional disparities and better inform us about the growing disease burden. To address these research gaps, we conducted a study to examine national trends in breast cancer incidence by race/ethnicity across stage, socioeconomic-position, and geography subgroups from 1999 to 2017.

MATERIALS AND METHODS

Study Population

We obtained cancer incidence data from the 1995-2018 NAACCR CiNA public-use data set (July 2021 release)[13] through the SEER*Stat program from the National Cancer Institute (NCI).[14] We analyzed data from 1999 to 2017 and excluded data from 2018 to limit the impact of case-reporting delays. We limited the analysis to available data from the United States within the 50 states or the District of Columbia; this included cancer registries participating in the National Program of Cancer Registries (the Centers for Disease Control and Prevention) and/or the NCI SEER program.[11] We excluded data from select states (Arkansas, Mississippi, Nevada, South Dakota, Tennessee, and Virginia) and the District of Columbia because they did not have data for all years from 1999 to 2017; this resulted in 44 states covering 92% of the US population.[15] We identified first primary invasive breast cancers, defined by codes C50.0 to C50.9 from the *International Classification of Diseases for Oncology, Third Edition*, among women aged 50 to 84 years to focus on breast cancer incidence rates among older women. We excluded women aged 85 years or older because that age group had no upper limit. We limited the study to non-Hispanic Black, non-Hispanic White, and Hispanic women to evaluate incidence rates by race/ethnicity (n = 2,726,922). We acknowledge the social construction of race/ethnicity and interpret race/ethnicity as a measure of sociopolitical experience.

We also accessed data on cancer stage (local, regional, or distant) based on SEER summary staging, the county-level percentages of persons living below the federal poverty line (<10%, 10%-19.99%, or ≥20%), the county-level urban/rural status, and the US Census geographic region. County-level poverty was defined by NAACCR with American Community Survey (ACS) 5-year estimates; however, because of data standards and confidentiality, it was not available before 2007. Thus, we used the ACS 2007-2011 categorization for the year 2007 and the ACS 2008-2012 and 2013-2017 categorizations for their respective years. The geographic region was categorized as West (Alaska, Arizona, California, Colorado, Hawaii, Idaho, Montana, New Mexico, Oregon, Utah, Washington, and Wyoming), Midwest (Illinois, Indiana, Iowa, Kansas, Michigan, Minnesota, Missouri, Nebraska, North Dakota, Ohio, and Wisconsin), South (Alabama, Delaware, Florida, Georgia, Kentucky, Louisiana, Maryland, North Carolina, Oklahoma, South Carolina, Texas, and West Virginia), or Northeast (Connecticut, Massachusetts, Maine, New Hampshire, New Jersey, New York, Pennsylvania, Rhode Island, and Vermont). The urban/rural status for a county from 2013 was defined by the US Department of Agriculture's rural-urban continuum codes with 2010 Census and 2006-2010 ACS data. A county was defined as urban (ie, metropolitan, codes 1-3) if it had 1 or more urbanized areas (dense urban entities with 50,000 or more people) or was an outlying county tied to a central metropolitan county according to the proportion of the labor force commuting. A county was defined as rural (ie, nonmetropolitan; codes 4-9, 88, or 99) if it was outside the metropolitan areas.[16] Rural counties are typically defined by open countryside, rural towns (less than 2500 people), and/or urban areas with 2500 to 49,999 people that are not part of metropolitan areas.[17]

Statistical Analysis

Using NCI SEER*Stat (version 8.3.8), we extracted age-adjusted (per the 2000 US standard population) incidence rates for breast cancer per 100,000 women, 95% confidence intervals (CIs), and standard errors for each year from 1999 to 2017 by a combination of race/ethnicity (Hispanic, non-Hispanic Black, and non-Hispanic White) and stage, county-level poverty, county-level urban/rural status, and US Census geographic region.

Using the NCI's Joinpoint Regression Program (version 4.7.0.0), we estimated longitudinal trends in age-adjusted breast cancer incidence rates from 1999 to 2017 by a combination of race/ethnicity and stage, county-level poverty, urban/rural status, and geographic region. The NCI's Joinpoint Regression Program uses joinpoint regression, also known as piecewise or segmented regression,

TABLE 1. Age-Adjusted Incidence Rates of Breast Cancer by Race/Ethnicity, Stage, Poverty, and Geography Among Women Aged 50 to 84 Years, North American Association of Central Cancer Registries, 2013-2017

	Incidence Rate (95% CI)[a]		
	Non-Hispanic Black	Non-Hispanic White	Hispanic
Overall	339.8 (337.3-342.3)	359.6 (358.7-360.6)	261.1 (258.9-263.3)
Stage at diagnosis			
Local	202.9 (201.0-204.9)	248.9 (248.0-249.7)	164.6 (162.8-166.4)
Regional	99.9 (98.6-101.3)	84.0 (83.5-84.5)	71.4 (70.2-72.5)
Distant	27.7 (27.0-28.4)	19.4 (19.2-19.7)	15.3 (14.8-15.9)
County-level % living in poverty			
<10%	355.3 (348.0-362.7)	381.3 (379.1-383.5)	280.8 (273.5-288.2)
10%-19.99%	339.4 (336.3-342.5)	356.1 (355.0-357.3)	260.0 (257.4-262.6)
≥20%	332.5 (327.3-337.8)	330.7 (327.5-334.0)	255.0 (249.5-260.6)
County-level urban/rural status			
Urban	340.8 (338.2-343.4)	366.3 (365.2-367.4)	262.7 (260.4-265.1)
Rural	326.1 (317.4-335.0)	329.1 (326.9-331.3)	234.4 (225.8-243.2)
Geographic region			
West	336.2 (328.0-344.6)	363.0 (360.9-365.2)	258.0 (254.4-261.6)
Midwest	355.9 (350.1-361.8)	359.3 (357.4-361.2)	241.8 (233.6-250.2)
South	336.4 (332.9-339.9)	346.1 (344.4-347.8)	259.0 (255.5-262.7)
Northeast	334.0 (328.4-339.6)	377.5 (375.3-379.6)	281.1 (275.4-286.9)

Abbreviation: CI, confidence interval.
[a]Incidence rate per 100,000 persons (age-adjusted to the 2000 US standard population).

to test for significant trends over time. We used a log-linear link for the regression model and set the outcome as the age-adjusted incidence rates and the year as the independent variable. We allowed for the program to estimate any joinpoints that suggested different trends over time. The joinpoint regression model first identifies an optimal number of joinpoints in the data by using a Monte Carlo permutation method.[18] Then, the model segments the entire time period by those join points and estimates an annual percent change for each segment. We also estimated the average annual percent change in the incidence rates across 1999 to 2017. We also estimated the average annual percent change across 2004 to 2017 to estimate a more recent trend after the decrease in breast cancer incidence rates attributed to the decline of hormone replacement therapy in the early 2000s.[6,19-21] When the optimal number of join points is zero, the annual percent change is equivalent to the average annual percent change. Tests for significance were 2-sided with P values < .05 considered statistically significant.

RESULTS

Among 2,726,922 women aged 50 to 84 years diagnosed with primary invasive breast cancer from 1999 to 2017, 10% were non-Hispanic Black, 83% were non-Hispanic White, and 7% were Hispanic. We report recent age-adjusted incidence rates and 95% CIs for breast cancer from 2013 to 2017 in Table 1. The age-adjusted incidence rates of breast cancer were 339.8 (95% CI, 337.3-342.3) per 100,000 persons for non-Hispanic Black women, 359.6 (95% CI, 358.7-360.6) per 100,000 persons for non-Hispanic White women, and 261.1 (95% CI, 258.9-263.3) per 100,000 persons for Hispanic women. We report the annual percent changes in the incidence rates from 1999 to 2017 in Table 2 and visualize the rates over time in Figure 1. In the text, we interpret only the recent trends from 2004 to 2017 and visualize those trends in Supporting Figure 1. We interpret trends that were not statistically different from 0 as stable over time and focus mainly on significant trends. From 2004 to 2017, the age-adjusted incidence rate increased annually by 0.9%, 0.4%, and 0.6% for non-Hispanic Black, non-Hispanic White, and Hispanic women, respectively.

Stage at Diagnosis

The age-adjusted incidence rates of local-stage breast cancer from 2013 to 2017 were highest for non-Hispanic White women, whereas the incidence rates of regional- and distant-stage breast cancer were highest for non-Hispanic Black women (Table 1). From 2004 to 2017, the average annual percent change in the age-adjusted incidence rate of local-stage breast cancer was 1.9% among non-Hispanic Black women, 1.2% among non-Hispanic White women, and 1.5% among Hispanic women (Table 2). In contrast, the age-adjusted incidence rate of regional-stage breast cancer decreased by 0.4%,

TABLE 2. Trends in Breast Cancer Incidence Rates by Race/Ethnicity Stratified by Stage, Poverty, and Geography Among Women Aged 50 to 84 Years, North American Association of Central Cancer Registries, 1999-2017

			APC							
			Trend 1		Trend 2		Trend 3		AAPC	
Strata	Race/Ethnicity	No.	Years	APC	Years	APC	Years	APC	2004-2017[a]	1999-2017
Overall	Non-Hispanic Black	267,942	1999-2005	−0.3	2005-2008	2.6	2008-2017	0.5[b]	0.9[b]	0.6
	Non-Hispanic White	2,275,555	1999-2001	−0.8	2001-2004	−4.1[b]	2004-2017	0.4[b]	0.4[b]	−0.5
	Hispanic	183,425	1999-2004	−1.6[b]	2004-2017	0.6[b]			0.6[b]	0
Stage										
Local	Non-Hispanic Black	149,488	1999-2004	−0.5	2004-2014	2.4[b]	2014-2017	0	1.9[b]	1.2[b]
	Non-Hispanic White	1,506,302	1999-2004	−3.6[b]	2004-2017	1.2[b]			1.2[b]	−0.2
	Hispanic	108,970	1999-2004	−1.8	2004-2017	1.5[b]			1.5[b]	0.6[b]
Regional	Non-Hispanic Black	86,035	1999-2008	1.1[b]	2008-2017	−1.0[b]			-0.4[b]	0
	Non-Hispanic White	582,458	1999-2017	−1.2[b]					-1.2[b]	−1.2[b]
	Hispanic	55,999	1999-2008	0	2008-2017	−1.4[b]			-1.0[b]	−0.7[b]
Distant	Non-Hispanic Black	21,900	1999-2008	2.8[b]	2008-2017	−0.1			0.8[b]	1.3[b]
	Non-Hispanic White	115,256	1999-2002	−1.9	2002-2014	1.9[b]	2014-2017	−2.1	0.9[b]	0.6
	Hispanic	10,479	1999-2017	0.9[b]					0.9[b]	0.9[b]
County-level % living in poverty[a]										
<10%	Non-Hispanic Black	21,423	2007-2017	0.8[b]					0.8[b]	
	Non-Hispanic White	283,591	2007-2017	0.4[b]					0.4[b]	
	Hispanic	13,353	2007-2017	0.4[b]					0.4[b]	
10%-19.99%	Non-Hispanic Black	116,192	2007-2017	0.4[b]					0.4[b]	
	Non-Hispanic White	938,409	2007-2017	0.3[b]					0.3[b]	
	Hispanic	92,936	2007-2017	0.6[b]					0.6[b]	
≥20%	Non-Hispanic Black	40,502	2007-2017	0.7[b]					0.7[b]	
	Non-Hispanic White	113,576	2007-2017	0.1					0.1	
	Hispanic	20,357	2007-2017	1.2[b]					1.2[b]	
County-level metropolitan status										
Urban	Non-Hispanic Black	247,219	1999-2005	−0.3	2005-2008	2.6	2008-2017	0.4[b]	0.9	0.5
	Non-Hispanic White	1,887,765	1999-2001	−0.9	2001-2004	−4.2[b]	2004-2017	0.4[b]	0.4[b]	−0.5
	Hispanic	173,091	1999-2004	−1.4	2004-2017	0.6[b]			0.6[b]	0
Rural	Non-Hispanic Black	20,645	1999-2017	1.1[b]					1.1[b]	1.1[b]
	Non-Hispanic White	387,114	1999-2005	−2.2[b]	2005-2017	0.5[b]			0.3[b]	−0.4[b]
	Hispanic	10,256	1999-2004	−4.6[b]	2004-2017	0.9[b]			0.9[b]	−0.7
Geographic region										
West	Non-Hispanic Black	25,208	1999-2005	−1.5	2005-2008	3.7	2008-2017	−0.7	0.3	−0.2
	Non-Hispanic White	478,150	1999-2004	−3.4[b]	2004-2017	−0.1			-0.1	−1.0[b]
	Hispanic	70,574	1999-2004	−1.6	2004-2017	0.8[b]			0.8[b]	0.1
Midwest	Non-Hispanic Black	56,562	1999-2004	−1.1	2004-2015	1.3[b]	2015-2017	−2.3	0.8	0.3
	Non-Hispanic White	604,489	1999-2001	−0.5	2001-2004	−4.7[b]	2004-2017	0.7[b]	0.7[b]	−0.3
	Hispanic	11,543	1999-2010	−1.0[b]	2010-2017	1.7[b]			0.5	0.1
South	Non-Hispanic Black	132,688	1999-2005	0	2005-2008	2.9	2008-2017	0.5[b]	1.0[b]	0.7[b]
	Non-Hispanic White	680,356	1999-2001	−0.2	2001-2004	−4.3[b]	2004-2017	0.4[b]	0.4[b]	−0.4
	Hispanic	68,734	1999-2004	−2.1	2004-2017	0.3			0.3	−0.4
Northeast	Non-Hispanic Black	53,484	1999-2004	−2.5[b]	2004-2017	0.6[b]			0.9[b]	0.9[b]
	Non-Hispanic White	512,560	1999-2017	0.9[b]					0.6[b]	−0.3[b]
	Hispanic	32,574	1999-2017	0.5[b]					0.5[b]	0.5[b]

Abbreviations: APC, annual percent change; AAPC, average annual percent change.
The color of the trend is grey for non-significant trends, blue for significant decreasing trends, and orange is for significant increasing trends.
[a]Trends for county-level poverty were available only from 2007 to 2017.
[b]Statistically significant with $P < .05$.

1.2%, and 1.0% annually from 2004 to 2017 for non-Hispanic Black, non-Hispanic White, and Hispanic women, respectively. The age-adjusted incidence rate of distant-stage breast cancer increased by 0.8% per year for non-Hispanic Black women and by 0.9% per year for non-Hispanic White and Hispanic women from 2004 to 2017.

County-Level Poverty

In counties with <10% or 10% to 19.99% poverty, the age-adjusted breast cancer incidence rates from 2013 to 2017 were highest among non-Hispanic White women. However, in counties with ≥20% poverty, the rates were highest among non-Hispanic Black women, but they were quite similar to those for non-Hispanic White

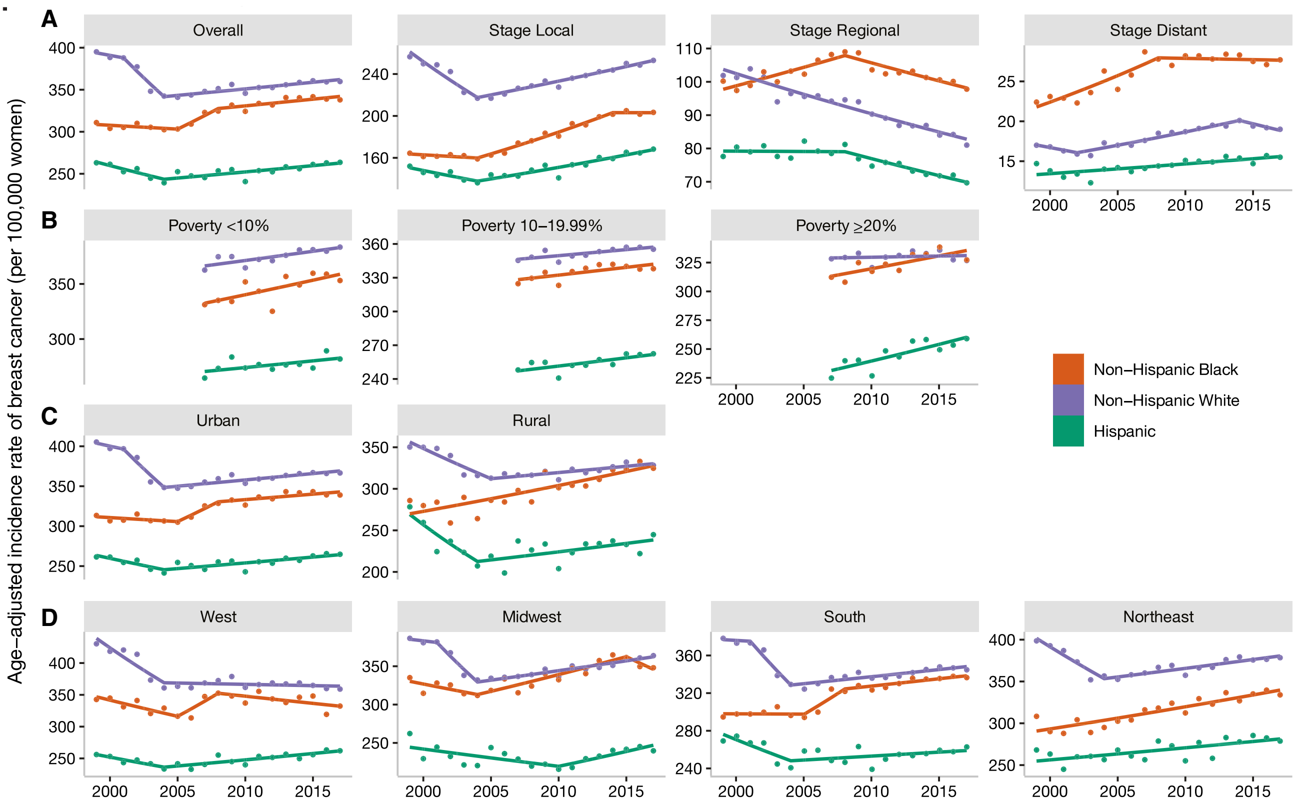

Figure 1. Age-adjusted incidence rates of breast cancer by race/ethnicity stratified by (A) stage, (B) poverty, (C) urban/rural status, and (D) region among women aged 50 to 84 years (North American Association of Central Cancer Registries, 1999-2017).

women (Table 1). Trends in breast cancer rates by poverty were available only from 2007 onward (Table 2). The age-adjusted incidence rate of breast cancer among counties with <10% poverty increased by 0.8% annually for non-Hispanic Black women and by 0.4% annually for non-Hispanic White and Hispanic women from 2007 to 2017. Among counties with 10% to 19.99% poverty, the age-adjusted incidence rate of breast cancer increased annually by 0.4% among non-Hispanic Black women, by 0.3% among non-Hispanic White women, and by 0.6% among Hispanic women from 2007 to 2017. Lastly, among counties with ≥20% poverty, the age-adjusted incidence rate of breast cancer increased annually by 0.7% among non-Hispanic Black women and by 1.2% among Hispanic women, whereas it was stable among non-Hispanic White women from 2007 to 2017.

The trends in local-stage breast cancer across race/ethnicity and poverty subgroups were similar to patterns in the overall trends, with larger trends found for non-Hispanic Black and Hispanic women in comparison with non-Hispanic White women (Fig. 2A and Supporting Table 1). Non-Hispanic Black women had the smallest decreasing trends for regional-stage disease across county-level poverty levels in comparison with non-Hispanic White and Hispanic women with 1 exception. In areas with ≥20% poverty, non-Hispanic Black and Hispanic women had the same decreasing trend in regional-stage breast cancer. For distant-stage disease, the only significant trends among poverty subgroups were increases among non-Hispanic White women in counties with 10% to 19.99% poverty and among non-Hispanic White and Hispanic women in areas with ≥20% poverty.

County Urban/Rural Status

In urban counties, the age-adjusted breast cancer incidence rates from 2013 to 2017 were highest for non-Hispanic White women (Table 1). In rural counties, the rates were similar for non-Hispanic Black and non-Hispanic White women. Among urban counties, from 2004 to 2007, the age-adjusted incidence rate of breast cancer increased by 0.4% and 0.6% per year among non-Hispanic White and Hispanic women, respectively (Table 2). The trend among non-Hispanic Black women in urban counties was stable; however, a later trend of 0.4% per year was

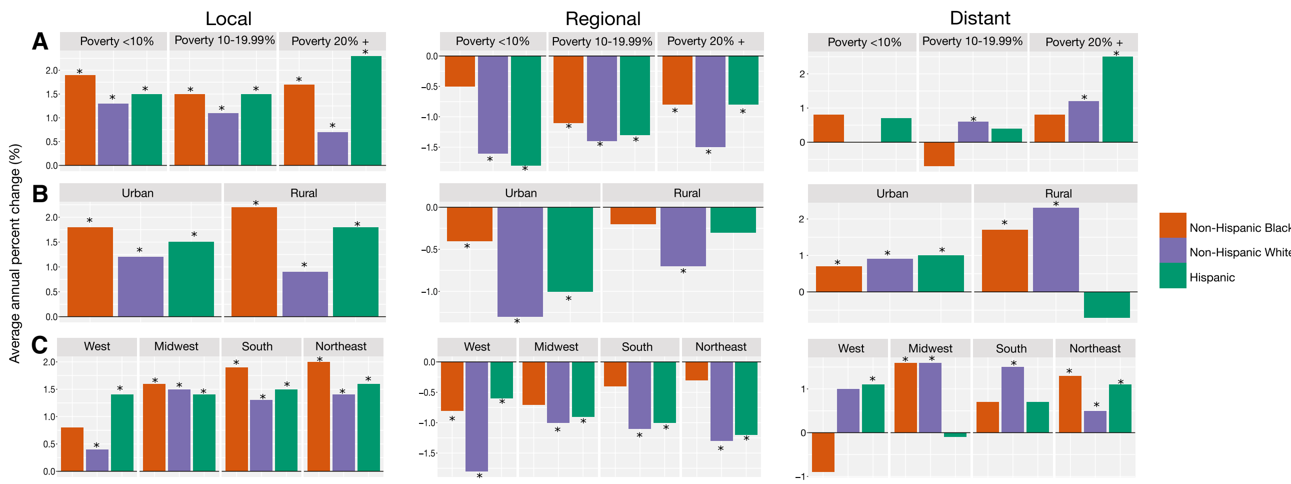

Figure 2. Average annual percent changes in the age-adjusted incidence rate of breast cancer from 2004 to 2017 by race/ethnicity and stage stratified by (A) poverty, (B) urban/rural status, and (C) region among women aged 50 to 84 years (North American Association of Central Cancer Registries, 1999-2017). Trends for county-level poverty were available only from 2007 to 2017. *Statistically significant (*P* < .05).

observed from 2008 to 2017. Among rural counties, the age-adjusted incidence rate increased by 1.1%, 0.3%, and 0.9% per year among non-Hispanic Black, non-Hispanic White, and Hispanic women, respectively.

The trends in local-stage breast cancer were similar to the overall trends, with larger increases found for non-Hispanic Black women in urban and rural counties in comparison with non-Hispanic White and Hispanic women (Fig. 2B and Supporting Table 1). Similarly, non-Hispanic Black women had the smallest declines in regional-stage disease in both urban and rural counties. For distant-stage disease, there were increasing trends in all race/ethnicity and urban/rural subgroups except for Hispanic women in rural areas, and the trends in rural areas were larger than the trends in urban areas.

Geographic Region

In the West, South, and Northeast, the age-adjusted incidence rates from 2013 to 2017 were highest for non-Hispanic White women (Table 1). In the Midwest, the age-adjusted incidence rate was highest among non-Hispanic White women, but it was similar to the rate among non-Hispanic Black women. The age-adjusted incidence rate of breast cancer increased by 0.8% per year from 2004 to 2017 among Hispanic women in the West (Table 2), and in supplemental analyses of the joint category of region and urban/rural status, the increase was limited to urban counties in the West (Supporting Table 2). In the Midwest, the age-adjusted incidence rate was stable for non-Hispanic Black and Hispanic women and increased by 0.7% per year among non-Hispanic White women from 2004 to 2017. The increase among non-Hispanic White women in the Midwest occurred in both urban and rural counties in the Midwest (Supporting Table 2). Additionally, there was a 1.1% annual increase in breast cancer from 2004 to 2017 among non-Hispanic Black women in urban counties in the Midwest that was not observed in the overall Midwest (Supporting Table 2). In the South, the age-adjusted incidence rate of breast cancer increased by 1% and 0.4% per year from 2004 to 2017 for non-Hispanic Black and non-Hispanic White women, respectively. The increase among non-Hispanic Black and non-Hispanic White women was observed in both urban and rural counties in the South (Supporting Table 2). In the Northeast, the age-adjusted incidence rate of breast cancer increased by 0.9%, 0.6%, and 0.5% per year from 2004 to 2017 among non-Hispanic Black, non-Hispanic White, and Hispanic women, respectively. The patterns for the Northeast overall were the same as the patterns for urban counties in the Northeast. No trends were observed in rural counties in the Northeast, although they could not be estimated for non-Hispanic Black and Hispanic women because of limited numbers (Supporting Table 2).

The trends for local-stage breast cancer were similar to the overall trends by geographic region (Fig. 2C and Supporting Table 2). For regional-stage disease, non-Hispanic Black women had the smallest declines in all regions except the West, where Hispanic women had the smallest decline. For distant-stage disease, there were significant increasing trends among Hispanic women in the West, among non-Hispanic Black and non-Hispanic White women in the Midwest, among non-Hispanic Black women in the South, and among all 3 racial/ethnic groups in the Northeast.

DISCUSSION

In this study of nationally representative data on breast cancer incidence rates from 1999 to 2017, we found increasing breast cancer incidence rates across race/ethnicity, county-level poverty, and geography, with stronger increases among non-Hispanic Black and Hispanic women compared with non-Hispanic White women. The stronger increases among non-Hispanic Blacks and Hispanics were particularly among those living in higher poverty areas versus lower poverty areas, rural areas versus urban areas, and most geographic regions. In our study, smaller trends in increasing breast cancer incidence rates were also observed for non-Hispanic White women in areas with <20% poverty, in urban and rural counties, and in most geographic regions. The trends of increasing breast cancer incidence rates across race/ethnicity as well as the increasing incidence of local- and distant-stage disease and decreasing incidence of regional-stage disease across race/ethnicity were similar to previous findings of studies that conducted analyses by stage.[1,5] However, in our study, the patterns were also observed in many urban/rural, poverty, and geographic subgroups that were not examined previously. The increasing incidence of breast cancer in all subgroups (especially distant-stage breast cancer) is concerning for the public health burden because the national cost of cancer care in the United States is highest for breast cancer (estimated to be $16.5 billion in direct medical spending in 2010).[22] Furthermore, disparities in the increasing breast cancer incidence across race/ethnicity and specific subgroups add additional concern.

Recent breast cancer incidence rates were highest among non-Hispanic White women and women living in urban or lower poverty areas, which should be targeted to reduce the breast cancer burden. However, stronger increases in breast cancer incidence were observed among non-Hispanic Black and Hispanic women living in counties with more poverty and in rural counties versus urban counties. The increasing rates raise several concerns for the burden of breast cancer among non-Hispanic Black and Hispanic women. Overall and within each stage, non-Hispanic Black women, followed by Hispanic women, have the worst breast cancer survival in comparison with non-Hispanic White women.[6] Women living in areas with greater poverty or a lower socioeconomic position have poorer breast cancer survival and greater mortality rates than their counterparts.[9,10,23] Furthermore, these studies have shown stronger disparities in breast cancer survival and mortality rates by socioeconomic position among rural areas and Black women.[23] Women living in rural areas are more likely to be diagnosed at a later stage[24] and less likely to receive screening mammography.[25] Breast cancer incidence increased among non-Hispanic Black women in the Northeast and South. The South also holds more non-Hispanic Black women with breast cancer than the other regions combined. Many states in the South have the highest breast cancer mortality rates and greatest breast cancer mortality disparity between Black and White women.[26] The growing breast cancer incidence rates may exacerbate the burden of breast cancer on the health system as well as the existing breast cancer mortality disparity for non-Hispanic Black women. The increasing breast cancer incidence among non-Hispanic Black women living in poverty, rural areas, and the South may be an important target to address the growing breast cancer burden.

We observed trends in increasing incidence rates for both local- and distant-stage breast cancer and declines in regional-stage breast cancer across race/ethnicity and among many subgroups. Although the decreasing breast cancer incidence rates in the early 2000s were attributed to the decline in the use of hormone replacement therapy,[6,19-21] the observed trends from 2004 to 2017 may be related to changes in breast cancer screening and the increasing prevalence of breast cancer risk factors. The decline in regional-stage disease is likely reflected by early detection due to improved screening access; however, from 2000 to 2015, the proportion of women reporting mammography screening declined among non-Hispanic Black, non-Hispanic White, and Hispanic women.[27] Thus, screening alone is unlikely to explain the increasing breast cancer incidence rates, especially for distant-stage disease. An ecologic study of breast cancer incidence rates among Black and White women in the United States from 1980 to 2011 evaluated trends in breast cancer risk factors and observed increases in body mass index and age at first live birth and decreases in age at menarche and the number of births per woman; these are trends that would increase breast cancer risk. The study found that increases in the average body mass index and parity among women in the United States were associated with the increasing incidence of breast cancer among both Black and White women.[28] Higher breast cancer incidence rates among non-Hispanic White women and women with a higher socioeconomic position have been associated with a higher prevalence of breast cancer risk factors, such as parity, age at first child birth, mammography utilization, and alcohol consumption.[29,30] In the ecologic study, the trends in breast cancer risk factors appeared to be stronger for Black women, and this could be related to the stronger increases in breast cancer incidence rates that we observed among non-Hispanic Black women and certain subgroups.[28] Reducing the prevalence of breast cancer risk factors is critical for reducing the rate of breast cancer and preventing growing disparities in the breast cancer burden. Among modifiable risk factors for postmenopausal breast cancer, weight change since the age of 18 years was estimated to be attributable to 18.7% of postmenopausal breast cancers, and it was followed by menopausal hormone therapy use (10.1%), alcohol consumption (5.9%), physical activity (3.3%), and breastfeeding (1.6%).[31] In fact, reducing these risk factors to the lowest risk levels could reduce the breast cancer incidence rate by a third.[31]

We leveraged population-based data from cancer registries to evaluate breast cancer incidence rates. The NAACCR data in this study cover more than 92% of the US population,[15] and this allows for greater geographic representation, sample sizes, and statistical power for analyses for subgroups in comparison with SEER, which covers a limited number of cancer registries in the United States. Because we limited our analyses to non-Hispanic Black, non-Hispanic White, and Hispanic women, we did not evaluate trends for Asian or American Indian/Alaska Native populations. We were limited by the data collected by cancer registries and area-level measures of socioeconomic position and thus were unable to evaluate socioeconomic position at the individual level. Additionally, delay-adjusted incidence rates were not available for the NAACCR public-use data set.

In conclusion, our study reveals greater increases in breast cancer incidence rates by race/ethnicity among poverty and geography subgroups. The increasing breast cancer incidence in areas with greater poverty and rural areas may exacerbate the breast cancer burden and existing disparities in breast cancer mortality, especially for non-Hispanic Black women. In light of these trends, additional efforts in breast cancer prevention through the reduction of modifiable breast cancer risk factors are critical for these disproportionately affected populations. Equally necessary are efforts to anticipate the growing breast cancer burden in these populations and ensure appropriate and regular screening and access to high-quality and timely care. These trends warrant further investigation and targeting for breast cancer prevention.

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CONFLICT OF INTEREST DISCLOSURES

The original analyses were conducted while Maneet Kaur was affiliated with Johns Hopkins; at the time of revisions, Maneet Kaur reported employment by Flatiron Health, Inc, an independent subsidiary of the Roche group, and stock ownership in Roche. The other authors made no disclosures.

AUTHOR CONTRIBUTIONS

Maneet Kaur: Design, analysis, and writing of the manuscript. **Corinne E. Joshu:** Critical feedback across all aspects of the manuscript. **Kala Visvanathan:** Critical feedback in the analysis and writing of the manuscript. **Avonne E. Connor:** Critical feedback across all aspects of the manuscript.

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